2.3 Assignment

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Machine Learning vs. Deep Learning: Applications in Customer Analytics

Introduction

Machine learning (ML) and deep learning (DL) are both important areas of artificial intelligence that allow computers to learn from data and make predictions. However, they differ in how they process information and identify patterns. Traditional machine learning often relies on human-selected features and algorithms such as logistic regression, decision trees, and support vector machines (SVM). Deep learning uses multi-layer neural networks to automatically identify more complex patterns, often from large amounts of raw data.

Both approaches have practical applications in business, especially in understanding customer behavior. Based on my professional experience in market research and analytics, I find these applications particularly interesting because companies increasingly rely on data to understand customers, improve products, and make strategic decisions. This report examines two examples: traditional machine learning for predicting customer churn in the telecommunications industry and deep learning for analyzing consumer feedback through natural language processing.

1. Traditional Machine Learning: Customer Churn Prediction

Real-world application

Customer churn prediction is an important application of machine learning in the telecommunications industry. Customer churn occurs when customers discontinue their subscriptions or switch to another service provider. Since acquiring new customers can be

expensive, identifying customers who may leave allows companies to develop retention strategies before losing them.

For example, a telecommunications company can use historical customer data to train a support vector machine (SVM) model to classify customers according to their likelihood of leaving. The model could use predefined features such as contract length, monthly charges, customer satisfaction, service usage, and subscription type. After learning from historical examples of customers who stayed or left, the model can evaluate existing customers and identify those who may be at risk of churning.

A real-world study by Amin et al. (2022) examined customer churn in the Malaysian telecommunications industry using 7,776 records from a telecommunications company's Net Promoter Score (NPS) dataset. The researchers compared several traditional machine learning techniques, including SVM, logistic regression, and decision trees, to identify factors associated with customer churn. Their findings highlighted the relationship between customer satisfaction and churn, demonstrating how machine learning can support customer retention strategies.

Why traditional machine learning is suitable

Traditional machine learning is suitable for this problem because customer churn prediction generally involves structured data with identifiable features. Information such as monthly charges, contract duration, and customer satisfaction scores can be organized into columns and used to train a classification model. SVM can learn patterns that distinguish customers who are likely to leave from those who are likely to stay.

Another advantage is that traditional machine learning can work effectively with relatively smaller datasets and requires less computational power than many deep learning models. It is also useful when analysts want to understand how specific customer characteristics relate to churn.

Deep learning would not necessarily be the best choice for this particular problem because the data is primarily structured and already contains identifiable features. A deep learning model could potentially identify more complex relationships, but it may require more data, computing resources, and time without providing enough additional value to justify the complexity.

2. Deep Learning: Natural Language Processing for Consumer Feedback

Real-world application

Another useful application of AI is analyzing customer feedback through natural language processing (NLP). Businesses collect large amounts of unstructured text through online reviews, surveys, customer service interactions, and social media. Reading and categorizing all these responses manually can be time-consuming, especially when companies receive thousands or even millions of comments.

Deep learning models, particularly Transformer models such as BERT, can help analyze this information by identifying context, meaning, and sentiment in written responses. For example, a company could use a Transformer-based model to classify customer feedback as positive, negative, or neutral while also identifying recurring topics such as product quality, pricing, customer service, and usability.

A 2025 study published in Scientific Reports introduced SentXFormer, a Transformer-based framework for sentiment analysis of customer reviews. The researchers evaluated the model using review datasets from Amazon, Yelp, and IMDb. The study demonstrated how Transformer-based deep learning models can identify sentiment across different types of customer feedback and perform well when analyzing reviews from different domains.

This application connects closely with my professional experience in market research. In my previous work, I supported consumer research projects that involved understanding customer preferences, evaluating product offerings, and turning research findings into recommendations. For instance, when analyzing survey responses, it is important to understand not only whether a customer likes or dislikes a product, but also the reasons behind that opinion. NLP tools could help researchers process large volumes of open-ended responses more efficiently and identify themes that may otherwise take significant time to uncover manually.

Why deep learning is suitable

Deep learning is suitable for NLP because language is complex and the meaning of a statement often depends on context. The same word or phrase can communicate different meanings depending on how it is used. Transformer models can analyze relationships between words and learn contextual representations from large amounts of text without requiring analysts to manually define every relevant feature.

Traditional machine learning could also be used for sentiment analysis. For example, an SVM model could classify customer comments using manually selected features such as keywords, word frequency, or sentiment indicators. However, this approach may struggle with sarcasm,

complicated sentence structures, and context-dependent meanings. It also requires more manual feature engineering.

Deep learning is therefore more suitable when a company has a large volume of unstructured text and needs to identify more complex language patterns. However, it also requires more computational resources and training data. For smaller datasets or relatively simple classification tasks, traditional machine learning may still be a practical and efficient option.

3. Comparison and Conclusion

Both traditional machine learning and deep learning can help businesses make better data-driven decisions, but their suitability depends on the type of problem and data being analyzed.

Comparison	Traditional ML: Churn Prediction	Deep Learning: Consumer Feedback
Data type	Structured customer data	Unstructured text
Example algorithm	Support Vector Machine (SVM)	Transformer / BERT
Feature extraction	Often requires human-selected features	Automatically learns complex features
Main purpose	Identify customers at risk of leaving	Understand sentiment and context in customer feedback
Main advantage	Efficient for structured data and smaller datasets	Handles complex language patterns

Main limitation	May struggle with complex relationships in raw text	Requires more data and computational resources

From my perspective, these two approaches serve different purposes rather than one being universally better than the other. In customer analytics, traditional machine learning can help identify patterns in structured information, such as customer spending and subscription behavior. Deep learning can provide additional value when analyzing more complex information, such as written customer feedback.

I also believe that combining these approaches can create opportunities for better business decisions. For example, a company could use traditional machine learning to identify customers who are likely to leave while using deep learning to analyze their feedback and better understand the reasons behind their dissatisfaction. Together, these insights could help businesses develop more targeted retention strategies and improve the customer experience.

Ultimately, choosing between machine learning and deep learning should depend on the business problem, the available data, and the resources required to implement the solution. Understanding these differences is important for selecting the right analytical approach and turning data into meaningful business recommendations.

References

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